

Triphala (PADMA) extract alleviates bronchial hyperreactivity in a mouse model through liver and spleen immune modulation and increased anti-oxidative effects

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Abstract:

Objectives: Triphala (TRP), a herbal extract from Tibetan medicine, has been shown to affect lymphocytes and natural killer T (NKT) cell function. We hypothesize that TRP could ameliorate bronchial hyperreactivity through immune-cell modulations.

Methods: Asthma mouse models were generated through intraperitoneal (IP) injections of ovalbumin (OVA)/2 weeks followed by repeated intranasal OVA challenges. Mice were then treated with normal saline (OVA/NS) or Triphala (OVA/TRP). Data were compared with mice treated with inhaled budesonide. All groups were assessed for allergen-induced hyperreactivity; lymphocytes from lungs, livers and spleens were analyzed for OVA-induced proliferation and their alterations were determined by flow cytometry. Oxidative reactivity using chemiluminescence, serum anti-OVA antibodies level and lung histology were assessed.

Results: Both TRP and budesonide significantly ameliorated functional and histological OVA-induced bronchial hyperreactivity. TRP had no effect on serum anti-OVA antibodies as compared with decreased levels following budesonide treatment. Furthermore, a significant increase in lung and spleen CD4 counts and a decrease in the liver were noted after TRP treatments. Bronchoalveolar fluid from TRP-treated animals but not from the budesonide-treated animals showed anti-oxidative effects.

Conclusion: TRP and budesonide caused a significant decrease in bronchial reactivity. TRP treatment altered immune-cell distributions and showed anti-oxidative properties. These findings suggest that immune-cell modulation with TRP can ameliorate lung injury.

Keywords: anti-oxidants, asthma, liver, lymphocytes, spleen

Introduction

The inflammatory process in asthma is mainly dominated by a Th-2 polarized cytokine profile [Busse and Lemanske, 2001; Cohn *et al.* 2004; Maddox and Schwartz, 2002; Grunig *et al.* 1998; Wills-Karp *et al.* 1998]. These inflammatory cells are detected in the bronchoalveolar lavage (BAL) of virtually all patients with asthma [Robinson *et al.* 1992] and are thought to play a pivotal role in bronchial hyperreactivity. The current paradigm of asthma is of imbalance between Th-1 and

Th-2 immune response, promoting increased inflammation in the airways which in turn causes increased airway hyperreactivity [Robinson *et al.* 1992; Kim *et al.* 2010]. Treatment aimed at reducing inflammation of the airways reduced airway hyperreactivity [Kim *et al.* 2010]. Moreover, recently natural killer T (NKT) cells were increasingly implicated in the pathogenesis of different disease processes [Yu and Porcelli, 2005], including bronchial asthma [Lisbonne *et al.* 2003; Akbari *et al.* 2003, 2006]. However, there is still

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controversy concerning the exact role of NKT cells in the pathogenesis of asthma.

Padma Hepaten is a formula derived from traditional Tibetan medicine [Tasduq *et al.* 2005, 2006; Naik *et al.* 2005] produced by Padma Inc. according to good manufacturing practice in Switzerland. It is a mixture of the medicinal plants *Terminalia chebula*, *Embica officinalis* and *Terminalia belerica*. This herbal mixture is rich in polyphenols, tannic acid and flavonoids. An important property of these three fruit preparations is its ability to scavenge reactive oxygen species [Ginsburg *et al.* 1999; Suter and Richter, 2000]. A similar formula of an equal proportional mixture is also known as Triphala (TRP), also recognized as Padma-Liver, P-Liver or Padma-28. These formulas have shown promising results in studies in atherosclerosis [Gieldanowski *et al.* 1992] as well as antiviral, antibacterial, anti-allergic, antimutagenic activities [Srikumar *et al.* 2006] and have radioprotective properties [Jageta *et al.* 2002].

TRP's mechanism of action is still poorly understood but it is in part attributed to its anti-oxidant properties [Ginsburg *et al.* 1999; Suter and Richter, 2000]. TRP has a specific influence on interferon- γ pathways [Neurauter *et al.* 2004] and alters the distribution of different lymphocyte subsets and decrease the number of NKT cells in the liver, in a murine model of hepatic fibrosis [Ginsburg *et al.* 2009]. These changes were associated with a significant amelioration of hepatic fibrogenesis and were not attributed to TRP's antioxidant properties. Nevertheless, recent reports have suggested that TRP can affect lymphocyte distribution and ameliorate inflammation in a hepatic fibrosis model regardless of its anti-oxidant properties [Ginsburg *et al.* 2009]. Therefore, the pattern of change in the lymphocyte distribution in the liver, spleen and lungs in an ovalbumin (OVA) model of lung injury can be better understood through the effects of TRP. We aimed in this study to determine the effect of TRP treatment on OVA-induced lung injury, in a murine model mimicking asthma [Horani *et al.* 2011].

Materials and methods

Animals

The study was approved by the animal ethics committee of the Hebrew University-Hadassah

Medical School. Male Balb/c mice, age 8–10 weeks, purchased and housed in a barrier facility and received care according to the National Institutes of Health guidelines. Animals were sacrificed 24 hours after the last administration of antigen. At the time of sacrifice, mice were weighed and anesthetized through the intraperitoneal (IP) injection of 0.1 ml of ketamine: xylazine:acepromazine (4:1:1) per 30 g of body weight.

Animal model of experimental allergic bronchitis

Experimental allergic bronchitis mimicking asthma was induced by IP injection of 10 μ g OVA (Grade III; Sigma-Aldrich) in 3 mg aluminum hydroxide (Al(OH)₃) on days 0, 7 and 14. Animals were thereafter challenged with intranasal instillation of OVA 100 μ g in 50 μ l saline, three times a week for 4 weeks starting on day 14 [Horani *et al.* 2011].

Experimental design

Following the OVA induction of asthma in three groups (10 animals each), mice were treated with IP injections of TRP. A naïve group receiving IP injections of saline served as healthy controls. The naïve group served as a negative control. Animals in this group were age matched with the other experimental groups and were housed in the same facility. These animals served as a quality control group to allow for controlling for changes that are not necessarily due to the treatments used. Data collected were compared with those from a fourth group that was treated with inhaled budesonide.

Treatments

Saline or TRP treatments were given every other day; starting on day 14, after asthma was induced. TRP was administered IP in a dose of 2 mg suspended in 200 μ l normal saline, per animal. Budesonide treatment was administered at a dose of 2 mg/5 ml/cage and administered using an inhaler directly to a sealed caged.

Bronchial allergen challenge assessment

Bronchial reaction to allergen inhalation was performed under continuous airflow conditions using a noninvasive method as described previously [Horani *et al.* 2011; Hamelmann *et al.* 1997; Lomask, 2006]. Briefly, bronchial allergen challenge assessment was performed after the

last allergen challenge. Mice were placed in a whole-body plethysmograph connected to a pneumotach, which was connected to a 10 ml bottle at its other end. The pneumotach was connected to a pre-amplifier. Analog signals from the amplifier were converted to digital signals by an AD card. Software (System XA, model SFT 1810, Buxco Electronics) was used to analyze 10 breath signals and calculate the plethysmographic enhanced pause (Penh) value. The Penh value is a unitless indicator of changes in airway resistance that correlates with specific airway resistance. It reflects the time of pressure decay to 36% of total box pressure during expiration. The Penh value was calculated before and 5 min after allergen challenge. Airway hyper-responsiveness was expressed as the percentage change of the Penh value. The Penh value was used as a noninvasive screening method to assess for changes in airway hyperresponsiveness in the studied animals.

Lymphocyte isolation from the liver and lung and splenocyte

All samples were isolated using the same protocols as described previously [Horani *et al.* 2011; Safadi *et al.* 2004]. Cells were counted and adjusted to 2×10^7 /ml in staining buffer (in saline containing 1% bovine serum albumin [BSA]). In brief, spleens were harvested at the time of sacrifice and fractionated through a 70- μ m nylon cell strainer. Red blood cells were lysed by the addition of lysis buffer containing NH_4Cl , KHCO_3 and Na-EDTA. Splenocytes were washed, suspended in RPMI 1640 medium and stored at 4°C until fluorescence-activated cell sorting (FACS) analysis. Livers and lungs were homogenized and incubated at 37°C for 30 min. Next, the digested liver or lung cell suspension was centrifuged to remove parenchymal cells and cell clumps. The supernatant was then centrifuged to obtain a pellet of cells depleted of parenchymal cells to a final volume of 1 ml. Lymphocytes were then isolated from this cell suspension by 24% metrizamide gradient cell separation.

Lung histology

Lungs were removed during sacrifice, and fixed by careful injection of paraformaldehyde into the main bronchus at a pressure of 20 cm H_2O . This allows for even inflation of the lung without undue distortion of the architecture. The middle third was embedded in paraffin and histology slides

were obtained randomly by longitudinal cut into three sections. Slides were stained with hematoxylin and eosin (H&E). The sections were reviewed by an experienced pathologist who was blind to the treatment groups.

Serum alanine aminotransferase

Blood samples were collected from the inferior vena cava and alanine aminotransferase (ALT) was measured using an automated enzymatic assay with the Vistros Chemistry Systems 950.

Flow cytometry

Lymphocytes were analyzed by fluorescence-activated cell sorter (FACScalibur, Becton Dickinson, Immunofluorometry systems, Mountain View, CA). Cells were passed at a rate of about 1000 per second, using saline as the sheath fluid. A 488 nm argon laser beam was used for excitation. Lymphocytes were identified and gated based on their size (forward light scatter [FSC]) and granularity (side light scatter [SSC]) and by staining with antibodies to CD45 specific for white blood cells (WBCs). Lymphocytes were stained with monoclonal antimouse CD4-FITC and CD8-PE and were detected by the FL-1 PMT and FL-2 PMT, respectively using log amplification. Natural killer (NK) cells were identified by staining with NK1.1 monoclonal antibody. For every assay, unstained cells, and anti-mouse-IgG negative controls were used. CD4, CD8, NK and NKT cell markers were chosen as general immune regulatory cell markers that are likely to change in response to increased inflammation, and to whether a Th-1 or Th-2 profile is dominant. These markers were used in conjunction with changes in immune globulin levels as well as lymphocyte proliferation potential to give an overview of the level of inflammation [Horani *et al.* 2011]. Instrument calibration and settings were performed using CaliBRITE™-3 beads (Becton Dickinson). The percentage of each antibody-attached cell population from the entire population of pan-leukocytes (CD45+) was calculated by the FACS-equipped CellQuest® software. NKT cells were identified by analyzing CD8/NK cells as well CD4/NK cells.

T-cell proliferation

Splenocytes were cultured in RPMI 1640 medium in the presence or absence followed by pulse with 1 μCi ^3H -thymidine as described earlier [Horani *et al.* 2011]. T-cell proliferation post

OVA stimulation was divided by that obtained from nonstimulated lymphocytes. The increased value was used to calculate the percentage of proliferation reduction post feeding [Safadi *et al.* 2005].

Anti-OVA serum antibody levels

Serum was separated from blood collected at the time of sacrifice and was evaluated for anti-OVA antibodies by enzyme-linked immunosorbent assay (ELISA) as described previously [Horani *et al.* 2011; Safadi *et al.* 2005].

Measurements of anti-oxidant capacities by a chemiluminescence-inducing cocktail

A Luminol-dependent chemiluminescence (LDCL) assay [Ginsburg *et al.* 2004a, 2004b] was employed to measure the oxidant scavenging abilities of whole mouse blood and of lavage samples. Samples were added to 800 μ l of Hanks' balanced salt solution which interacted with a flux of oxidants generated by a cocktail composed of Luminol (10 μ M), glucose oxidase (GO) from *Aspergillus niger* (1.46 units), as a source of H_2O_2 , 2 mM sodium selenite (IV) (a source of electrons) and 10 μ M of Cobalt (II). This cocktail generates a constant flux of H_2O_2 and of hydroxyl radical and a steep peak of LDCL. We mixed 10 μ l BAL and 10 μ l of plasma with the cocktail, vortexed briefly and the degree of light quenching was monitored for several minutes. Light generation, which persisted for about 10 min at a high steady level, was measured in a LUMAC type 2500 luminometer attached to a PC computer (Landgraaf, the Netherlands). This assay measures the ability of the anti-oxidants to quench luminescence generated by the oxidants from the cocktail.

Statistical analysis

The FACS and aminotransferase results levels were analyzed with analysis of variance (ANOVA) and Student's *t*-test. An ANOVA with Dunnett's test for multiple comparisons was used to determine the statistical significance of differences from the values obtained for nonsensitized animals after OVA challenges. A two-way ANOVA was used with Tukey's multiple range tests to determine the statistical significance of differences among strains after OVA challenges.

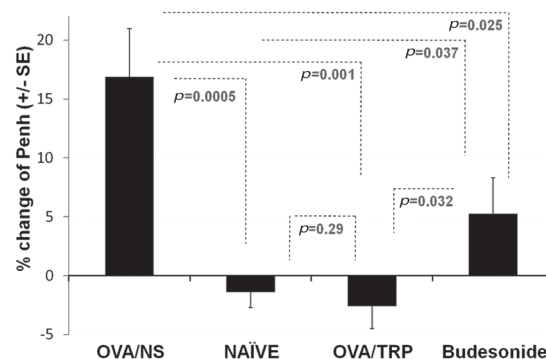


Figure 1. Airway reactivity expressed as the percentage change of Penh. Airway reactivity was evaluated as the percentage change of Penh ($\pm$ SE) using a whole-body plethysmograph. Both TRP and budesonide treatments significantly decreased the percentage change of Penh. The TRP effect was more prominent than budesonide ($p = 0.03$).

Results

Treatment with TRP, following asthma induction, significantly decreased the percentage change of Penh (enhanced pause)

Airway hyperreactivity was measured using a whole-body plethysmograph and expressed as the percentage change of Penh $\pm$ SEM (Figure 1). Penh was $-1.38 \pm 1.3\%$ in the naïve state and significantly increased to $16.86 \pm 4.1\%$ ($p = 0.0005$) following asthma induction that was treated by normal saline. Treatment with IP TRP significantly decreased the change of Penh to $-2.59 \pm 1.9\%$ ($p = 0.001$) and to $5.26 \pm 3.1\%$ ($p = 0.025$) with budesonide. The Penh score of the treatment group was not significantly different from the naïve group ($p = 0.29$). The improvement seen after treatment with TRP was even better than that seen after treatment with inhaled budesonide ($p = 0.032$). Animals treated with IP TRP showed no obvious differences in behavior or weight as compared with the other treatment groups.

TRP-attenuated lung inflammation

OVA-induced lung injury was associated with abundant inflammatory infiltrations, as evident in the H&E-stained lung sections (Figure 2). Prominent peribronchial and perivascular infiltrations were seen after OVA-induced lung injury (Figure 2b) as compared with the naïve group (Figure 2a). The inflammatory infiltrations were clearly attenuated by treatment with

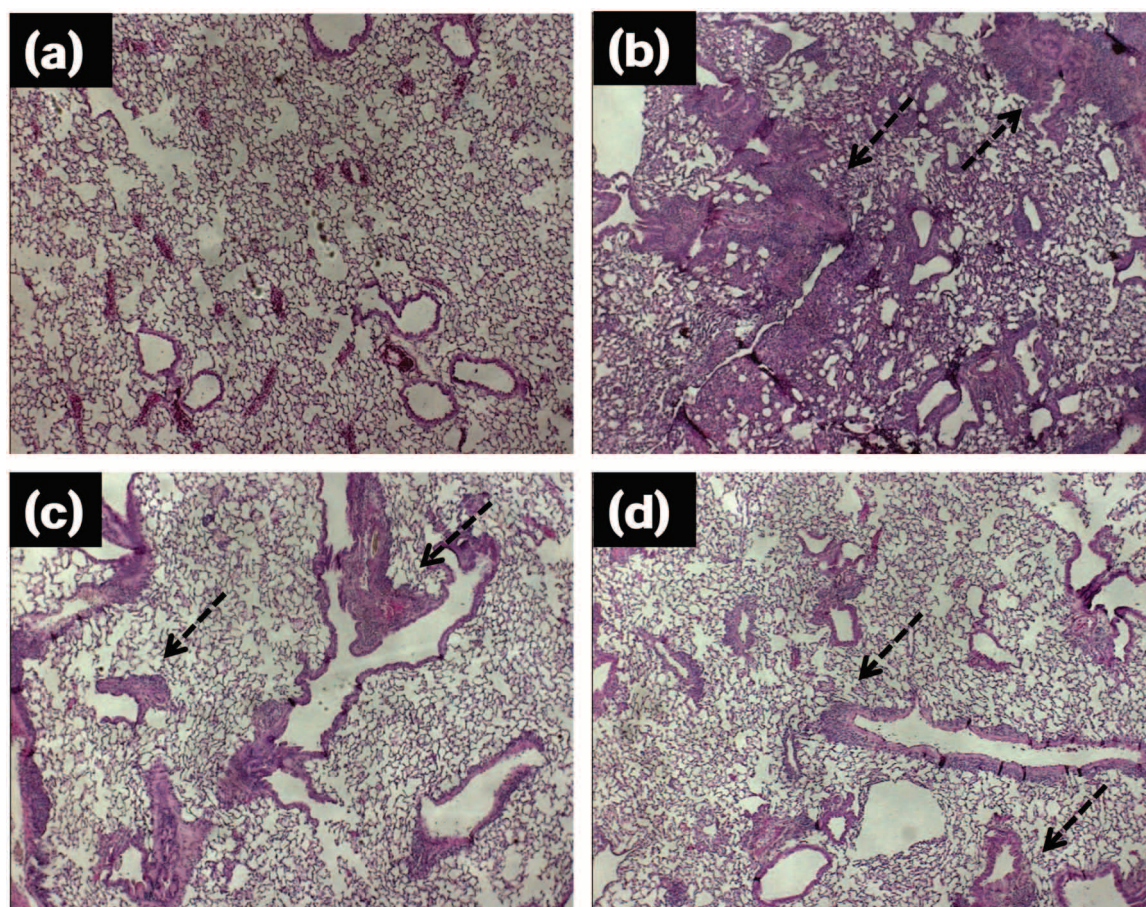


Figure 2. Hematoxylin and eosin (H&E) stain of lung tissue. The middle third of the lung section was embedded in paraffin and histology slides were obtained randomly by longitudinal cutting into three. Slides were stained with H&E. Following staining, $\times 20$ images were acquired. Compared with the naïve group (a), prominent perivascular and peribronchial infiltration are seen in ovalbumin (OVA)-induced asthma (b). These infiltrations were attenuated in the budesonide (c) and in the Triphala (TRP) (d) groups.

IP TRP (Figure 2c) which was comparable to that seen after treatment with inhaled budesonide (Figure 2d).

The effect of TRP in modulating OVA-induced lung injury was not mediated through a humoral response

Semiquantitative ELISA was performed to study the serum level of anti-OVA antibodies at different dilutions. Arbitrary primary ELISA optical density (OD) readings were used to assess anti-OVA titers in a semiquantitative manner (Figure 3).

Serum anti-OVA titers at the 1:5000 dilutions were 0.512 ± 0.06 OD in the presence of naïve untreated serum. Anti-OVA serum titers rose significantly ($p < 0.0001$) to 1.376 ± 0.04 in the untreated OVA-induced group when compared

with the naïve group. Treatment of OVA-induced animals with TRP had no effect on the high serum anti-OVA titers, which remained high, similar to the untreated group (1.459 ± 0.06 , $p = 0.16$) and significantly higher than the levels in sera from the naïve group ($p < 0.0001$). Treatment of OVA-induced animals with budesonide, as expected, caused a significant decrease in the anti-OVA titer to 0.693 ± 0.11 OD. This observed level was similar to that observed in non-OVA-induced naïve animals ($p = 0.12$) suggesting an efficient humoral suppression of anti-OVA antibodies production.

TRP did not suppress T-cell proliferation

We reported earlier no budesonide effect on T-cell proliferation derived splenocytes in the OVA-induced lung injury [Horani *et al.* 2011]. In order

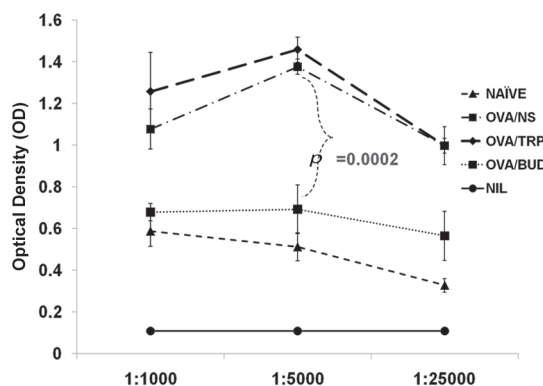


Figure 3. Semiquantitative enzyme-linked immunosorbent assay (ELISA) analysis of anti-ovalbumin (OVA) antibodies. Serum anti-OVA titers were measured by ELISA using arbitrary OD units and presented with means $\pm$ standard error. Increased serum anti-OVA titers, expressed as increased OD readings, express successful induction of asthma in this model. A significant fall in serum anti-OVA levels was achieved by the budesonide treatment as compared with the untreated group ($p = 0.0002$). The beneficial effect of Triphala (TRP) did not affect production of anti-OVA antibodies when compared with the untreated OVA group.

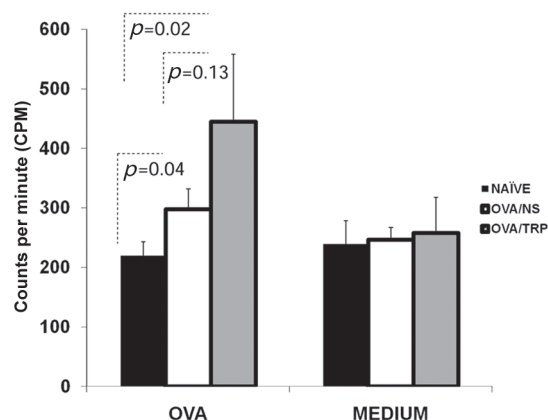


Figure 4. T-cell proliferation post ovalbumin (OVA) stimulation. Spleen T cells from all groups showed low and almost similar proliferative profiles when cultured in medium only (right panel). Specific stimulation with OVA, however, caused significant increased proliferation in all cells originally coming from OVA-treated donors but not from naïve group (left panel). Triphala (TRP) treatment caused a nonsignificant trend of increase in T-cell proliferation compared with budesonide ($p = 0.13$).

to study the effect of TRP on the cellular response, T-cell proliferation was measured in splenocytes before and after TRP treatment (Figure 4). Spleen T cells from all study groups showed low and similar proliferative profiles when cultured in medium only (Figure 4, right panel). T-cell proliferation cultured in medium was 239.25 ± 39.6 counts per minute (cpm) in naïves, 249 ± 20.3 cpm in asthma-induced animals and 257 ± 60.2 cpm in the TRP-treated animals. Specific stimulation with OVA increased proliferation in both asthmatic groups but not in the cells from the naïve group (Figure 4, left panel). T-cell proliferation following OVA stimulation was 220 ± 23 cpm in naïves which increased to 297.5 ± 34.7 cpm in untreated asthma ($p = 0.04$) and 444.5 ± 113.8 cpm in the TRP-treated group ($p = 0.02$), respectively. Although TRP caused increased proliferation of lymphocytes compared with the untreated OVA group, this increase was not statistically significant.

TRP affected immune cell distribution

FACS was performed on splenocytes as well as isolated liver and lungs lymphocytes of the different study groups. We previously reported that treatment with budesonide had no significant

changes on lung, liver or spleen NKT cells when compared with an untreated OVA-induced asthma group. Similarly, budesonide treatment did not cause significant change in CD4+ cell counts in either the lung or liver compared with an untreated OVA-induced asthma group. Spleen CD4 cell count, however, increased significantly after budesonide treatment [Horani *et al.* 2011]. We only show here the result of TRP therapy. The results of the different lymphocyte subsets are presented as the percentage of all CD45+ cells $\pm$ SEM (Figure 5).

There was no significant change in NKT levels in the lung (NK+CD3+) following asthma induction in either treated or untreated groups. Furthermore, we did not find a significant change in spleen NKT cell numbers after asthma induction. No significant differences were seen among the different OVA treatment groups.

Liver NKT cells, showed a decrease in NKT cell numbers after OVA induction, yet this change barely reached significance. Liver NKT cell numbers were $1.65 \pm 0.42\%$ in the naïve study group, which decreased to $1.02 \pm 0.12\%$ and $0.87 \pm 0.15\%$ in the nontreated OVA and TRP-treated group ($p = 0.14$ and $p = 0.07$, respectively).

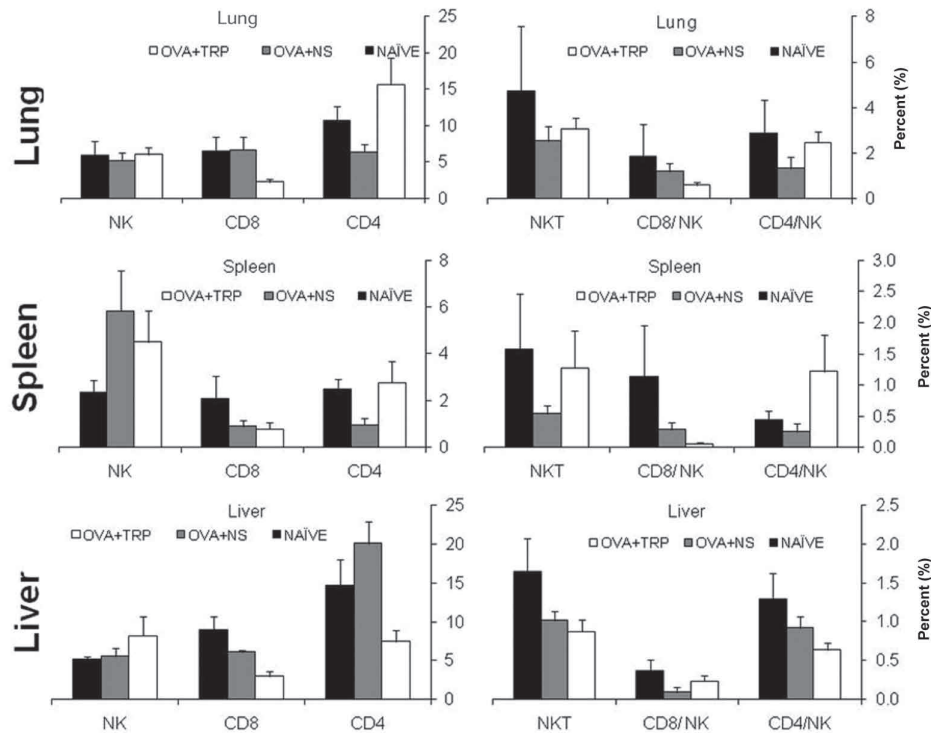


Figure 5. Flow cytometry analysis of liver, spleen and lung lymphocytes. Isolated lymphocytes from each organ were stained as described in the methods section. Triphala (TRP) treatment caused a significant increase in lung CD4 counts paralleling a significant decrease in the liver. Furthermore, TRP treatment caused a decrease in lung CD8 cells.

When studying NK cell numbers, we only found significant changes in spleen NK cell numbers (NK+ CD3-). Following OVA induction, spleen NK cells increased from $2.32 \pm 0.55\%$ in naïve animals to $5.83 \pm 1.7\%$ after OVA induction ($p = 0.02$). Yet, there was no significant difference between the untreated and treated groups ($p = 0.3$).

When studying lung CD4+ cells distribution, we noticed a decrease in CD4+ cell numbers following OVA induction as compared with naïve non-treated animals ($10.67 \pm 1.93\%$ to $6.39 \pm 1.06\%$, $p = 0.07$). TRP treatment on the other hand caused a significant increase in CD4 cell numbers to $15.66 \pm 3.58\%$ ($p = 0.1$ and $p = 0.03$ when compared with naïves and the nontreated OVA group, respectively).

Spleen CD4+ cells showed a similar trend to that seen in the lungs, such that CD4+ numbers significantly decreased following OVA induction and increased following treatment with TRP. CD4 cell numbers were $2.48 \pm 0.41\%$ in the naïve group as compared with $0.96 \pm 0.29\%$ in the untreated

asthma group ($p = 0.01$) and $2.76 \pm 0.91\%$ in the TRP-treated group ($p = 0.4$ and $p = 0.07$ when compared with the naïve and OVA groups, respectively).

Following OVA induction, liver CD4+ cells increased from $14.66 \pm 3.36\%$ as seen in the naïve group to $20.12 \pm 2.74\%$ in the untreated asthma group, but this increase did not reach statistical significance ($p = 0.17$). TRP treatment on the other hand, caused a significant decrease in CD4+ cell numbers to $7.5 \pm 1.49\%$ ($p = 0.05$ and $p = 0.04$ compared with the naïve and OVA non-treated groups, respectively).

Examining CD8+ cell distribution, we noted no significant change in lung CD8+ numbers after OVA induction ($6.5 \pm 1.97\%$ in naïve group and $6.7 \pm 1.78\%$ in the OVA untreated group, $p = 0.4$). TRP treatment, on the other hand, caused a significant decrease in lung CD8+ cells to $2.28 \pm 0.3\%$ ($p = 0.04$ and $p = 0.01$ when comparing naïve and TRP treatment groups and OVA untreated and TRP treatment groups, respectively). Unlike the lungs, OVA induction caused a

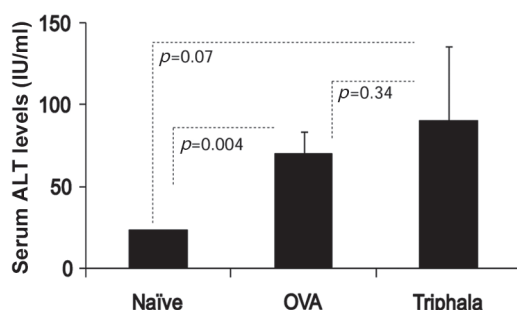


Figure 6. Serum alanine aminotransferase (ALT) levels determination. Serum ALT levels increased significantly after asthma induction ($p = 0.004$).

significant decrease in liver CD8+ numbers from $8.95 \pm 1.76\%$ in the naïve group to $6.1 \pm 0.17\%$ and $3.01 \pm 0.65\%$ in the untreated and TRP-treated groups ($p = 0.04$ and $p = 0.008$, respectively). There was no significant difference in liver CD8+ cells between the OVA-induced groups. No significant changes were seen in CD8+ cells numbers in the spleen.

In summary, TRP treatment caused a significant increase in lung CD4 numbers paralleling a significant decrease in the liver. Furthermore, TRP treatment caused a decrease in lung CD8 cells. These data might suggest that TRP had an effect on the lungs through an effect on lymphocyte distribution.

OVA induction caused increased ALT levels

We also investigated liver injury reflected by levels of serum ALT, as a screening tool for liver involvement in our model. Serum ALT levels, increased from 22.96 ± 1 IU/ml in naïves to 69.68 ± 14.5 IU/ml in the OVA-induced asthma mice and 90.03 ± 45.9 IU/ml in the TRP treatment group ($p = 0.004$ and $p = 0.07$, respectively). There was no significant difference between the untreated asthma groups and the TRP-treated group (Figure 6).

TRP had effect on the BAL but not on blood anti-oxidant activity

The anti-oxidant capacity of blood and of lung lavage was assessed using the LDCL assay (as described in the methods section). The intensity of light readings are presented as mean values and standard error (Figure 7). The intensity of luminescence correlated linearly with the generation of radicals, and anti-oxidants therefore resulted in

a decline of radicals and decreased intensity of luminescence. The intensity of light, generated by the GO cocktail alone (not shown), was at a constant level of approximately 252,000 cpm. BAL obtained from all animal groups (Figure 7a) showed a maintained increase of anti-oxidant activity as compared with naïve animals. This anti-oxidant activity became significant ($p < 0.04$) in the TRP treatment group compared with the control OVA-induced asthma starting at 360 seconds. There were no significant differences in between other groups up to 540 seconds of *in vitro* follow up. After 60 seconds, blood samples from all groups (Figure 7b) achieved a similar peak of anti-oxidant activity (250,000 arbitrary units) as compared with the lavage samples. However, serum anti-oxidant activity dropped later on, but without significant differences between the different groups.

Discussion

The current study demonstrates that TRP, an herbal blend derived from traditional Tibetan medicine, alleviates bronchial reactivity in a murine model of asthma (Figure 1). The effects of TRP were compared with budesonide, a well-established treatment for asthma that has become a cornerstone for asthma control. Inhaled corticosteroids allow for decreased airway inflammation and thus decreased airway hyperreactivity [McMillan *et al.* 2005]. Pulmonary Penh value improvement observed with TRP treatment was even superior to the classical benefit achieved by budesonide. TRP treatment achieved a significant decrease in inflammation as seen in H&E sections (Figure 2). The exact mechanism of TRP in different disease entities is not entirely elucidated. The beneficial effect of TRP has been attributed at least in part to its antioxidant properties [Ginsburg *et al.* 1999; Mahesh *et al.* 2009]. In the current study, TRP was associated with increased anti-oxidative activity in the lavage but not serum samples as measured by the luminescence cocktail (Figure 7). The negligible anti-oxidative activity in serum samples compared with that in whole blood was recently discussed [Ginsburg *et al.* 2012].

TRP treatment had no significant effect on *in vitro* OVA-stimulated T-cell proliferation (Figure 4), suggesting a different pathway in alleviating asthma-associated inflammation not necessarily through a direct inhibition on T-cell proliferation. Airflow obstruction in asthma is associated with airway

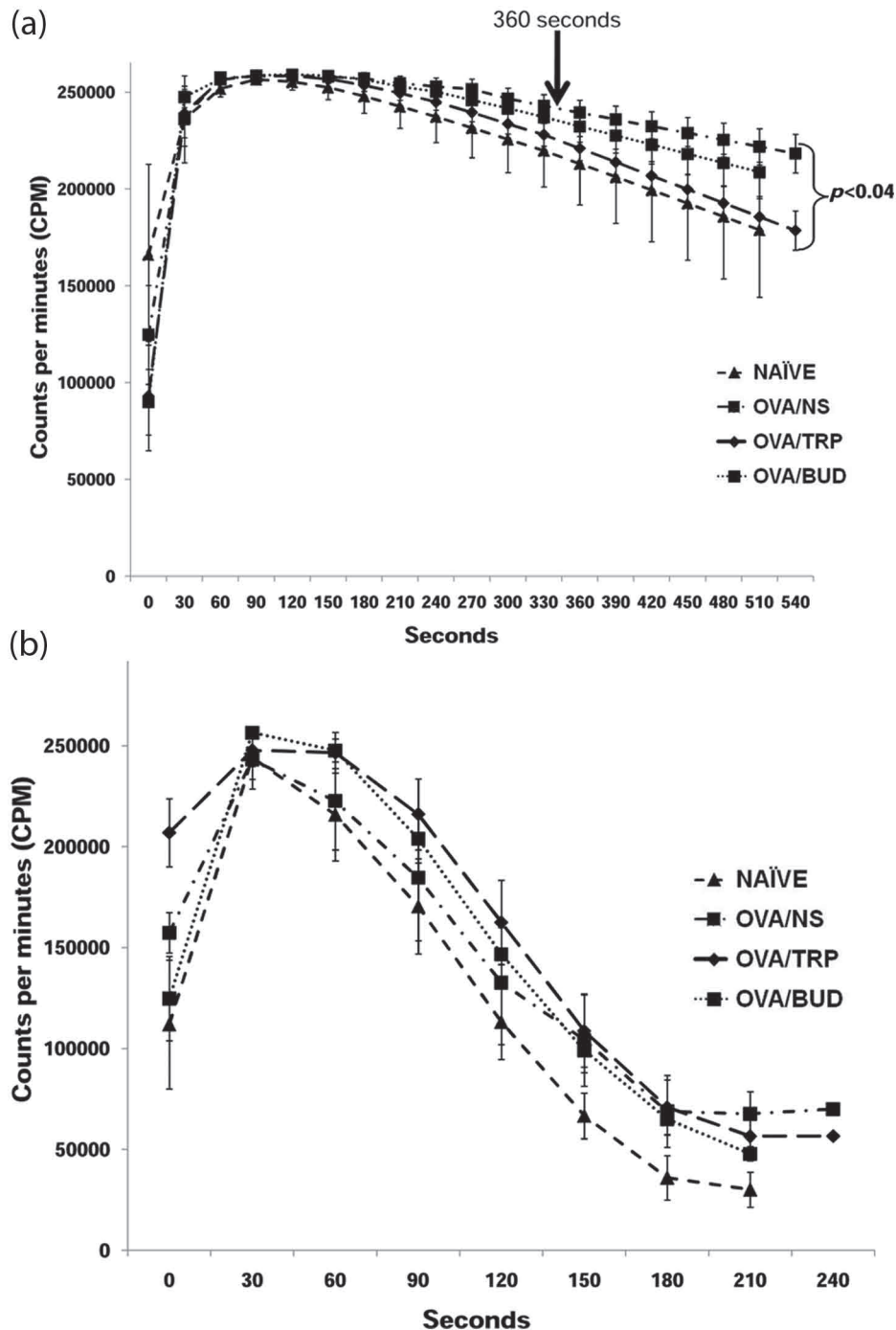


Figure 7. Triphala (TRP) had no affect on the bronchoalveolar lavage (BAL) or but not blood anti-oxidant activity. A Luminol-dependent chemiluminescence (LDCL) induced by the GO [glucose oxidase] cocktail is shown, using lavage (a) and whole blood (b) samples obtained from naïve animals, untreated ovalbumin (OVA)-induced lung injury, TRP- and budesonide-treated animals with OVA lung injury. The intensity of light, generated by the GO cocktail alone (not shown) was at a constant level of 252,000 cpm. Data represent mean $\pm$ standard error (SE) from six animals per group. (a) Luminol-dependent chemiluminescence (LDCL) induced by the GO cocktail, using 10 μ l amounts of plasma. (b) Lavage samples. This anti-oxidant activity was significant ($p < 0.04$) in the BAL of the TRP treatment group compared to the control OVA induced asthma starting at 360 seconds. There were no significant differences in the plasma anti-oxidant activity between the different groups.

inflammation that is enriched with eosinophils and CD4⁺ T lymphocytes [Busse and Lemanske, 2001]. The role of CD4⁺ T cells in the pathogenesis of asthma is well documented [Bosse and Rola-Pleszczynski, 2007]. The paralleled decrease in CD4⁺ cells in the liver as a result of TRP exposure can be the result of extrapulmonary immune organs being a target for timing, migration and location of antigen-specific T-cell division during airway inflammation [Hutchison *et al.* 2009]. Nevertheless, recent reports have suggested that TRP can affect lymphocytes distribution and ameliorate inflammation in a hepatic fibrosis model regardless of its anti-oxidant properties [Ginsburg *et al.* 2009]. Other than lungs, in the current study we have investigated liver and spleen lymphocyte alterations, for possible distinct phases and locations of antigen presentation in airways inflammation. In our model of lung injury, TRP treatment induced a significant multi-organ shift in CD4⁺ cells distribution (Figure 5). Following asthma induction, lung CD4⁺ T cells decreased. TRP treatment, on the other hand, caused a significant increase in lung CD4⁺ cells. This latter increase was paralleled by a significant decrease in liver CD4⁺ content. The elevation of CD4⁺ T cells with the protected phenotype observed in the TRP-treated mice could be reconcile with T regulatory cells or perhaps T cells that secrete high levels of IL-10 or other anti-inflammatory mediators.

NKT cells play an essential pathogenic role in asthma [Akbari *et al.* 2006]. Activation of NKT cells can either induce or prevent allergen-induced airway hyperreactivity [Hachem *et al.* 2005; Stock *et al.* 2004; Kim *et al.* 2004]. Although TRP was previously shown to significantly affect hepatic NKT cells in a model of liver fibrosis [Ginsburg *et al.* 2009], we did not see significant changes in NKT cell numbers in the lungs, livers or spleen after IP administration of TRP. That said, we did notice a decreasing trend in NKT cell numbers in the liver and an increasing trend in the lungs. The overall pattern of change in the lymphocyte distribution in the liver, spleen and lungs suggests a possible immune regulation in the OVA model of lung injury through an effect on lung CD4 and possibly NKT lymphocytes, which may further explain the beneficial effect of TRP. The induction of adaptive immunity requires antigen-presenting cells (APCs), and dendritic cells (DCs) are the main types of APCs involved in the induction of TH2 responses to allergens in asthma. In the lung, DCs can be found throughout the conducting airways, interstitium, vasculature and

pleura and in bronchial lymph nodes [Kim *et al.* 2010]. In addition, the imbalance between Th-1 and Th-2 immune response in asthma, promote increased inflammation in the airways which in turn causes increased airway hyperreactivity [Robinson *et al.* 1992; Kim *et al.* 2010].

We recently suggested a multi-organ shift of lymphocyte subsets in a murine model of OVA-induced lung injury mimicking asthma [Horani *et al.* 2011]. Lymphocytes 'redistribution' in the liver, lung and spleen was associated with decreased lung reactivity as well as decreased lymphocytic infiltration seen after treatment with β -glucosylceramide. This observation is further strengthened by the relative hepatotoxicity seen after OVA induction (Figure 6). Serum ALT levels confirm liver involvement as they increased following OVA induction in untreated and, to a lesser extent, TRP-treated groups with lung injury.

TRP's effect on the adoptive immune response seems to be different than the anti-humoral effect of budesonide (Figure 3). Anti-OVA antibodies further confirm that budesonide alleviates OVA-induced asthma via an effect on the humoral arm of the immune system, compared with TRP that had no effect on antibody production.

Triphala's beneficial effect in a murine model of asthma can be caused by both an immunomodulatory effect as well as an anti-oxidative effect working in concert. Indeed the importance of increased oxidative burden in the pathogenesis of asthma is increasingly recognized [Yalcin *et al.* 2012; Hox *et al.* 2011; Celik *et al.* 2012] suggesting that lowering the oxidative injury in asthmatic lungs can potentially lower airway hyperactivity. We have previously shown that Triphala's immunomodulatory effects are independent of its anti-oxidant properties [Horani *et al.* 2011] suggesting that both the immunomodulation and anti-oxidative capacity observed in the bronchoalveolar fluid may indeed be synergic.

In conclusion, TRP treatment caused significant amelioration of OVA-induced asthma. This amelioration was associated with changes in the distribution of lymphocyte subsets in lungs, liver and spleen. TRP was also associated with increased anti-oxidative activity in the lavage samples. Our data extend our understanding of the effect of lymphocyte alterations in asthma and the potential role of safe natural compounds in asthma treatment.

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Conflict of interest statement

All authors declare that there are no conflicts of interest

References

- Akbari, O., Faul, J., Hoyte, E., Berry, G., Wahlstrom, J., Kronenberg, M. *et al.* (2006) CD4⁺ invariant T-cell-receptor⁺ natural killer T cells in bronchial asthma. *N Engl J Med* 354: 1117–1129.
- Akbari, O., Stock, P., DeKruyff, R. and Umetsu, D. (2003) Role of regulatory T cells in allergy and asthma. *Curr Opin Immunol* 15: 627–633.
- Bosse, Y. and Rola-Pleszczynski, M. (2007) Controversy surrounding the increased expression of TGF beta 1 in asthma. *Respir Res* 8: 66–70.
- Busse, W. and Lemanske, R., Jr (2001) Asthma. *N Engl J Med* 344: 350–362.
- Celik, M., Tuncer, A., Soyer, O., Saçkesen, C., Tanju Besler, H. and Kalayci, O. (2012) Oxidative stress in the airways of children with asthma and allergic rhinitis. *Pediatric Allergy Immunol*, in press.
- Cohn, L., Elias, J. and Chupp, G. (2004) Asthma: mechanisms of disease persistence and progression. *Annu Rev Immunol* 22: 789–815.
- Gieldanowski, J., Dutkiewicz, T., Samochowiec, L. and Wojcicki, J. (1992) Padma 28 modifies immunological functions in experimental atherosclerosis in rabbits. *Arch Immunol Ther Exp* 40: 291–295.
- Ginsburg, I., Koren, E., Horani, A., Mahamid, M., Doron, S., Muhanna, N. *et al.* (2009) Amelioration of hepatic fibrosis via Padma Hepaten is associated with altered natural killer T lymphocytes. *Clin Exp Immunol* 157: 155–164.
- Ginsburg, I., Kohen, R. and Koren, E. (2012) Is the exclusive use of plasma to quantify oxidant-scavenging abilities in clinical settings representing the true redox status in vivo? *Engl J Med*, in press.
- Ginsburg, I., Sadovnic, M., Oron, M. and Kohen, R. (2004a) Novel chemiluminescence-inducing cocktails, part I: the role in light emission of combinations of luminal with SIN-1, selenite, albumin, glucose oxidase and Co2+. *Inflammopharmacology* 12: 289–303.
- Ginsburg, I., Sadovnic, M., Oron, M. and Kohen, R. (2004b) Novel chemiluminescence-inducing cocktails, part II: measurement of the anti-oxidant capacity of vitamins, thiols, body fluids, alcoholic beverages and edible oils. *Inflammopharmacology* 12: 305–320.
- Ginsburg, I., Sadovnik, M., Sallon, S., Milo-Goldzweig, I., Mechoulam, R., Breuer, A. *et al.* (1999) PADMA-28, a traditional Tibetan herbal preparation inhibits the respiratory burst in human neutrophils, the killing of epithelial cells by mixtures of oxidants and pro-inflammatory agonists and peroxidation of lipids. *Inflammopharmacology* 7: 47–62.
- Grunig, G., Warnock, M., Wakil, A., Venkayya, R., Brombacher, F., Rennick, D. *et al.* (1998) Requirement for IL-13 independently of IL-4 in experimental asthma. *Science* 282: 2261–2263.
- Hachem, P., Lisbonne, M., Michel, M., Diem, S., Roongapinun, S., Lefort, J. *et al.* (2005) Alpha-galactosylceramide-induced iNKT cells suppress experimental allergic asthma in sensitized mice: role of IFN-gamma. *Eur J Immunol* 35: 2793–2802.
- Hamelmann, E., Schwarze, J., Takeda, K., Oshiba, A., Larsen, G., Irvin, C. *et al.* (1997) Noninvasive measurement of airway responsiveness in allergic mice using barometric plethysmography. *Am J Respir Crit Care Med* 156: 766–775.
- Horani, A., Shoseyov, D., Doron, S., Mruwat, R., Amer, J., Kerem, E. *et al.* (2011) Immune modulation of ovalbumin-induced lung injury in mice using beta-glucosylceramide and a potential role of the liver. *Immunobiology* 216: 548–557.
- Hox, V., Vanoirbeek, J., Callebaut, I., Bobic, S., De Vooght, V., Ceuppens, J. *et al.* (2011) Airway exposure to hypochlorite prior to ovalbumin induces airway hyperreactivity without evidence for allergic sensitization. *Toxicol Lett* 204: 101–107.
- Hutchison, S., Choo-Kang, B., Gibson, V., Bundick, R., Leishman, A., Brewer, J. *et al.* (2009) An investigation of the impact of the location and timing of antigen-specific T cell division on airways inflammation. *Clin Exp Immunol* 155: 107–116.
- Jagetia, G., Baliga, M., Malagi, K. and Sethukumar Kamath, M. (2002) The evaluation of the radioprotective effect of Triphala (an ayurvedic rejuvenating drug) in the mice exposed to gamma-radiation. *Phytomedicine* 9: 99–108.
- Kim, H., DeKruyff, R. and Umetsu, D. (2010) The many paths to asthma: phenotype shaped by innate and adaptive immunity. *Nat Immunol* 11(7): 577–584.
- Kim, J., Kim, D., Chang, W., Hong, C., Park, S., Kim, S. *et al.* (2004) Asthma is induced by intranasal

- coadministration of allergen and natural killer T-cell ligand in a mouse model. *J Allergy Clin Immunol* 114: 1332–1338.
- Lisbonne, M., Diem, S., de Castro Keller, A., Lefort, J., Araujo, L., Hachem, P. *et al.* (2003) Cutting edge: invariant V alpha 14 NKT cells are required for allergen-induced airway inflammation and hyperreactivity in an experimental asthma model. *J Immunol* 171: 1637–1641.
- Lomask, M. (2006) Further exploration of the Penh parameter. *Exp Toxicol Pathol* 57(Suppl. 2): 13–20.
- Maddox, L. and Schwartz, D. (2002) The pathophysiology of asthma. *Annu Rev Med* 53: 477–498.
- Mahesh, R., Bhuvana, S. and Begum, V. (2009) Effect of *Terminalia chebula* aqueous extract on oxidative stress and antioxidant status in the liver and kidney of young and aged rats. *Cell Biochem Funct* 27: 358–363.
- McMillan, S., Xanthou, G. and Lloyd, C. (2005) Therapeutic administration of Budesonide ameliorates allergen-induced airway remodelling. *Clin Exp Allergy* 35: 388–396.
- Naik, G., Priyadarsini, K., Bhagirathi, R., Mishra, B., Mishra, K., Banavalikar, M. *et al.* (2005) In vitro antioxidant studies and free radical reactions of triphala, an ayurvedic formulation and its constituents. *Phytother Res* 19: 582–586.
- Neurauter, G., Wirleitner, B., Schroecksnadel, K., Schennach, H., Ueberall, F. and Fuchs, D. (2004) PADMA 28 modulates interferon-gamma-induced tryptophan degradation and neopterin production in human PBMC in vitro. *Int Immunopharmacol* 4: 833–839.
- Robinson, D., Hamid, Q., Ying, S., Tsicopoulos, A., Barkans, J., Bentley, A. *et al.* (1992) Predominant TH2-like bronchoalveolar T-lymphocyte population in atopic asthma. *N Engl J Med* 326: 298–304.
- Safadi, R., Alvarez, C., Ohta, M., Brimnes, J., Kraus, T., Mehal, W. *et al.* (2005) Enhanced oral tolerance in transgenic mice with hepatocyte secretion of IL-10. *J Immunol* 175: 3577–3583.
- Safadi, R., Ohta, M., Alvarez, C., Fiel, M., Bansal, M., Mehal, W. *et al.* (2004) Immune stimulation of hepatic fibrogenesis by CD8 cells and attenuation by transgenic interleukin-10 from hepatocytes. *Gastroenterology* 127: 870–882.
- Srikumar, R., Parthasarathy, N., Manikandan, S., Narayanan, G. and Sheeladevi, R. (2006) Effect of Triphala on oxidative stress and on cell-mediated immune response against noise stress in rats. *Mol Cell Biochem* 283: 67–74.
- Stock, P., Kallinich, T., Akbari, O., Quarcoo, D., Gerhold, K., Wahn, U. *et al.* (2004) CD8(+) T cells regulate immune responses in a murine model of allergen-induced sensitization and airway inflammation. *Eur J Immunol* 34: 1817–1827.
- Suter, M. and Richter, C. (2000) Anti- and pro-oxidative properties of PADMA 28, a Tibetan herbal formulation. *Redox Rep* 5: 17–22.
- Tasduq, S., Mondhe, D., Gupta, D., Baleshwar, M. and Johri, R. (2005) Reversal of fibrogenic events in liver by *Emblica officinalis* (fruit), an Indian natural drug. *Biol Pharmaceut Bull* 28: 1304–1306.
- Tasduq, S., Singh, K., Satti, N., Gupta, D., Suri, K. and Johri, R. (2006) *Terminalia chebula* (fruit) prevents liver toxicity caused by sub-chronic administration of rifampicin, isoniazid and pyrazinamide in combination. *Human Exp Toxicol* 25: 111–118.
- Wills-Karp, M., Luyimbazi, J., Xu, X., Schofield, B., Neben, T., Karp, C. *et al.* (1998) Interleukin-13: central mediator of allergic asthma. *Science* 282: 2258–2261.
- Yalcin, A., Gorczynski, R., Parlak, G., Kargi, A., Bisgin, A., Sahin, E. *et al.* (2012) Total antioxidant capacity, hydrogen peroxide, malondialdehyde and total nitric oxide concentrations in patients with severe persistent allergic asthma: its relation to omalizumab treatment. *Clin Lab* 58: 89–96.
- Yu, K. and Porcelli, S. (2005) The diverse functions of CD1d-restricted NKT cells and their potential for immunotherapy. *Immunol Lett* 100: 42–55.